InTraScale: Accelerating Video Generation via In-Trajectory Resolution Scaling

Anonymous submission

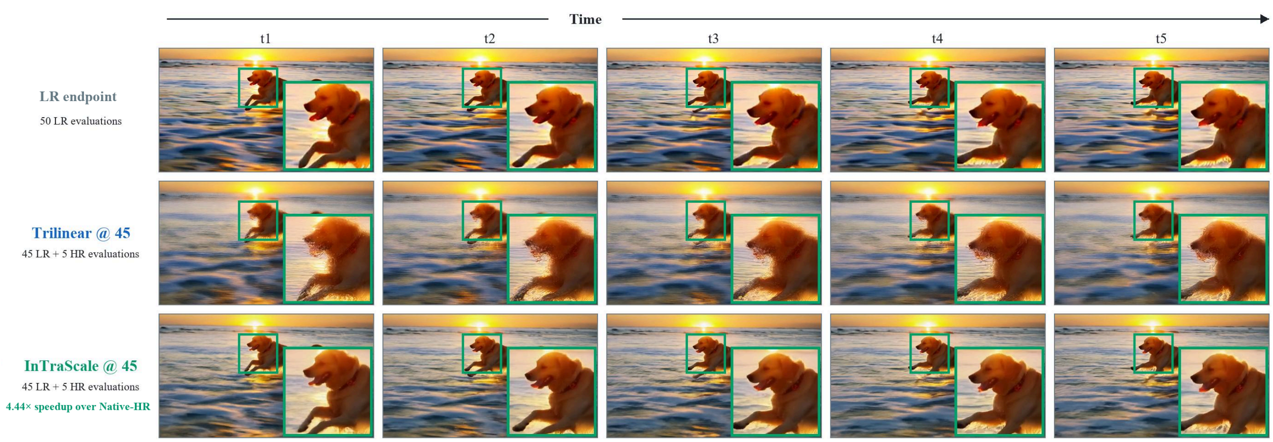

Figure 1: **In-trajectory resolution scaling for efficient high-resolution video generation.** Five matched frames compare the completed LR endpoint after 50 low-resolution evaluations, Trilinear@45, and InTraScale@45. With the same 45-evaluation low-resolution prefix, trilinear lifting causes severe detail degradation, whereas InTraScale preserves the content and temporal structure of the LR endpoint with only five target-resolution evaluations, achieving a $4.44\times$ speedup over Native-HR.

Abstract

Iterative denoising over long spatiotemporal sequences has become the primary performance bottleneck in high-resolution video diffusion. We define *in-trajectory resolution scaling* as a single sampling process that executes an initial denoising prefix at low resolution, transforms the intermediate latent state to the target resolution, and completes the remaining denoising steps at the target resolution. By allocating a fixed denoising-step budget across resolutions, this paradigm reduces target-resolution computation without invoking a separate super-resolution diffusion process or increasing the total denoising-step budget. However, *interpolation-based* methods cannot recover the correspondence between different resolution latents, while *upsampler-based* solutions face a training-inference mismatch between completed clean latents and predictions from an unfinished trajectory. To address these problems, we introduce **InTraScale**, a decoupled framework that addresses these challenges while keeping the pretrained generator frozen. First, **Trajectory-Tail Distillation (TTD)** distills the omitted low-resolution trajectory tail and calibrates the transition clean prediction toward the completed low-resolution endpoint. Then, the **In-Trajectory Upsampler (ITU)**, trained on paired low- and target-resolution clean latents, maps the calibrated prediction into the target-resolution latent space. A scheduler-consistent state reconstruction then returns the prediction to the remaining target-resolution trajectory. On 50-step Wan2.1 model, InTraScale achieves $4.44\times$ speedups while retaining 97.53% of the Native-HR VBench-5 score. On distilled 4-step Wan2.1 model, it achieves a $2.13\times$ speedup over Native-HR while retaining 99.58% of the native distilled score. These results establish in-trajectory resolution scaling as an effective spatial acceleration strategy complementary to timestep distillation.

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

1 Introduction

Recent video diffusion models generate increasingly coherent motion and fine-grained visual content, but their computational cost remains a major obstacle to practical deployment (Wan et al. 2025). Each denoising step applies a large diffusion transformer to the complete spatiotemporal latent sequence, making target-resolution generation particularly expensive. Existing acceleration techniques reduce this cost from different and largely orthogonal dimensions, including timestep distillation, sparse attention, and quantization. Spatial acceleration provides another complementary direction: it follows a low-to-high-resolution pipeline that establishes global layout, semantic structure, and motion on a compact latent grid, while reserving target-resolution computation for local structure and high-frequency details.

Such low-to-high-resolution pipelines can follow two schedules, depending on whether resolution is increased after low-resolution generation or before its trajectory is complete. Most existing pipelines follow the former, completing a low-resolution trajectory before applying an RGB-space or latent-space upsampling stage (Ho et al. 2022; Blattmann et al. 2023; Wang et al. 2023b). RGB-space cascades incur decode–encode and scheduler-state reconstruction overhead, whereas latent cascades directly upsample the completed representation (Xie et al. 2025; Zhao et al. 2026). Both, however, perform resolution enhancement only after low-resolution generation has completed. In-trajectory resolution scaling follows the latter: it transitions earlier and uses the remaining denoising steps as the target-resolution continuation (Jeong et al. 2025; ModelTC 2026), avoiding a separate super-resolution diffusion process or an increase in the original step budget.

Moving the resolution transition inside the denoising trajectory, however, introduces two additional mismatches. First, fixed latent interpolation changes the spatial grid without modeling the VAE-specific correspondence between low- and target-resolution representations. Second, a learned upsampler trained on completed clean latents must operate on a clean prediction produced by an unfinished low-resolution trajectory. We examine their technical implications in Section 3.1.

To address these problems, we introduce **InTraScale**, a decoupled framework for learned in-trajectory resolution scaling. Rather than requiring a single transition module to simultaneously correct the unfinished trajectory and reconstruct the target-resolution representation, InTraScale assigns these functions to independently trained components. The **In-Trajectory Upsampler (ITU)** learns the VAE-specific correspondence between clean low- and target-resolution latent pairs. **Trajectory-Tail Distillation (TTD)** adapts the transition prediction before upsampling by distilling the omitted low-resolution trajectory tail into an update at the transition step. This separation allows ITU to focus on cross-resolution latent lifting, while TTD handles the state mismatch introduced by applying in-trajectory scaling.

At inference time, InTraScale first executes an inexpensive low-resolution denoising prefix. At the selected transition step, TTD produces an endpoint-calibrated low-resolution clean prediction, which ITU then maps to the target-resolution latent space. Then, a parameter-free scheduler-consistent reconstruction converts it into the corresponding target-resolution state. The frozen flow predictor then executes the remaining denoising steps at the target resolution. The resulting pipeline retains a fixed total denoising-step budget and requires neither a separate super-resolution diffusion process nor the VAE decode-encode costs.

Our contributions are summarized as follows:

- We introduce InTraScale, a learned in-trajectory resolution-scaling framework that distributes a fixed denoising-step budget between an inexpensive low-resolution prefix and a target-resolution continuation.
- We formulate the in-trajectory transition as a decoupled sequence of trajectory-state calibration, VAE-specific latent lifting, and scheduler-consistent state reconstruction, with all learned components supervised by the frozen base generator.
- We evaluate InTraScale on conventional 50-step and distilled 4-step Wan2.1 samplers, demonstrating substantial spatial acceleration both before and after timestep distillation.

2 Related Work

Video Diffusion Models. Latent video diffusion performs generation in compressed spatiotemporal representations (Blattmann et al. 2023); recent systems combine flow-matching objectives with DiT backbones (Jin et al. 2025; Wan et al. 2025), but high-resolution video still induces long latent sequences and costly attention and feed-forward computation.

Efficient Video Diffusion. Existing acceleration methods target several complementary dimensions. Timestep and consistency distillation shorten the sampling trajectory (Wang et al. 2023a; Li et al. 2024; Ding et al. 2025; Nie et al. 2026). Sparse attention reduces token interactions (Xi et al. 2025), while caching reuses attention maps or intermediate features (Zhao et al. 2025b; Liu et al. 2025; Zou et al. 2026). Quantization lowers the arithmetic and memory cost of DiT inference (Zhao et al. 2025a), and dynamic-computation methods vary model capacity or patch size across steps and regions (Anagnostidis et al. 2025; Kim, Ghadiyaram, and Gadde 2026; Li et al. 2026). These directions act on step count, operator cost, or feature reuse and can therefore be layered with spatial acceleration; our experiments verify this compatibility for timestep distillation.

Resolution-Scaling Video Generation. *Post-generation methods* enhance a completed low-resolution result in RGB space (Zhou et al. 2024; Xu et al. 2024; He et al. 2024) or latent space (Xie et al. 2025; Zhao et al. 2026); related systems append high-resolution refinement or distill restoration into a separate stage (Zheng et al. 2026b; Zhuang et al. 2026). *In-trajectory methods* instead vary resolution within the generative process. Some train the generator around multiscale trajectories (Gu et al. 2024; Jin et al. 2025), whereas methods for pretrained models prescribe variable-resolution schedules or expand noisy states during sampling (Jeong et al. 2025; ModelTC 2026; Zheng et al. 2026a; Xiao et al. 2026). InTraScale belongs to the latter category but keeps the generator frozen, learns the VAE-specific correspondence between clean low- and target-resolution latents, and separately corrects the unfinished low-resolution trajectory tail before scheduler-consistent target-resolution continuation.

3 Method

This section develops InTraScale by decomposing the transition into cross-resolution reconstruction and unfinished-

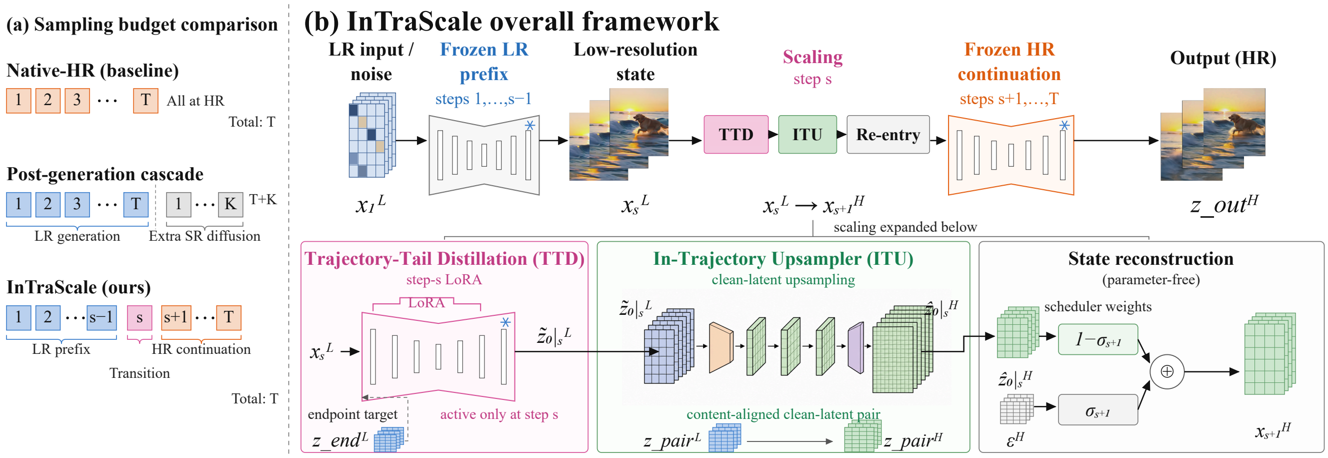

Figure 2: **Sampling budget and overall framework of InTraScale.** (a) Native-HR executes all T steps at target resolution, whereas a post-generation cascade appends K super-resolution steps after completing low-resolution generation. (b) At step s, TTD distills the omitted low-resolution trajectory tail into an endpoint-calibrated clean prediction. ITU, trained on low- and target-resolution clean-latent pairs, lifts this prediction to the target-resolution latent space.

trajectory correction. Accordingly, ITU learns the clean-latent correspondence across resolutions, TTD calibrates the transition prediction toward the completed low-resolution endpoint, and parameter-free state reconstruction returns the lifted latent to the target-resolution trajectory.

3.1 Challenges of In-Trajectory Transition

We now examine the two transition mismatches outlined in the Introduction and formalize why they require a decoupled transition design.

Latent distortion from trilinear interpolation. A fixed trilinear interpolation operator only resamples the spatial grid; it does not recover the correspondence between the low- and target-resolution representations induced by the pretrained VAE. Consequently, interpolated latents may exhibit distorted structures and inaccurate high-frequency statistics on the target-resolution grid. RALU reported that late latent upsampling introduces aliasing artifacts, particularly in high-frequency regions around object boundaries (Jeong et al. 2025). This exposes a fundamental quality–efficiency trade-off: a later resolution transition provides greater acceleration but leaves fewer target-resolution steps to correct the errors introduced during resolution conversion. With only a short target-resolution suffix, the remaining target-resolution steps may therefore be insufficient to repair structural distortions and synthesize details absent from the low-resolution representation.

Training–inference mismatch for learned upsampling. Replacing interpolation with a learned latent upsampler introduces an input distribution shift. A latent upsampler is naturally trained using paired clean latents obtained from completed low- and target-resolution videos. Its low-resolution training input is thus a completed clean representation containing all spatial and temporal details supported by the low-resolution bandwidth. During in-trajectory inference, however, the upsampler does not receive such a converged latent. Let x_s^L denote the input to transition step s, the last step executed at low resolution; steps $s+1, \dots, T$ run at target resolution. Here, v_ϕ is the frozen flow predictor with parameters ϕ, c is the text condition, and σ_s is the scheduler coefficient. We use the subscript $0|s$ for the clean latent predicted at the transition step. The upsampler therefore receives a transition clean prediction derived from an unfinished low-resolution trajectory:

$$\hat{z}_{0|s}^L = x_s^L - \sigma_s v_\phi(x_s^L, s, c).$$

It contains a timestep- and scheduler-dependent residual that would otherwise be reduced by the remaining low-resolution steps. Its distribution therefore differs from that of the clean low-resolution latents used to train the upsampler. Directly applying the upsampler to this prediction forces a single module to perform both cross-resolution reconstruction and unfinished trajectory correction, weakening its ability to preserve structure across scales.

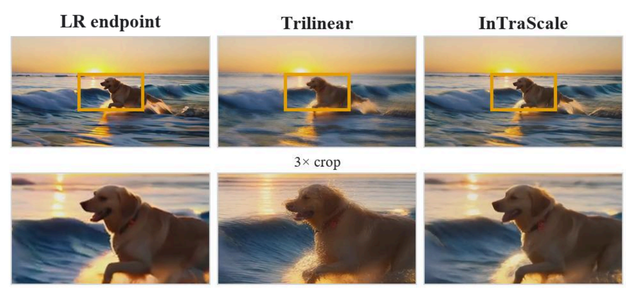

Figure 3: Trilinear interpolation under a late resolution transition. The columns show the completed LR endpoint after 50 low-resolution evaluations, Trilinear@45, and InTraScale@45; the bottom row shows the corresponding 3× crops.

Figure 4: Comparison of endpoint output after step 50, unfinished-trajectory residual at transition step 45, and TTD-calibrated output at transition step 45

Together, these mismatches motivate the overall structure of InTraScale. We next formalize the resulting InTraScale pipeline.

3.2 Overall Framework

Figure 2 summarizes the sampling budget and the decoupled pipeline motivated by the two challenges above. We now formalize its components and trajectory states. Let $v_\phi(x_k^R, k, c)$ denote the frozen flow predictor, where $R \in \{L, H\}$ indicates the low- or target-resolution latent space, x_k^R is the latent state presented at denoising step k, and c is the text condition. We use "step" to denote one scheduler update and its corresponding model evaluation, σ_k for the scheduler coefficient, and T for the total number of denoising steps. We refer to the pretrained flow predictor and VAE collectively as the frozen base generator. Following the convention above, step s is the transition step and the last step executed on the low-resolution grid. Steps $1, \ldots, s-1$ form the frozen low-resolution prefix, the TTD update is applied only at step s, and steps $s+1, \ldots, T$ form the frozen target-resolution continuation.

At the transition step, the prediction $\widehat{z}^L_{0|s}$ introduced in the preceding subsection is not directly passed to the upsampler. Trajectory-Tail Distillation (TTD) first produces an endpoint-calibrated clean prediction $\widetilde{z}^L_{0|s}$. The In-Trajectory Upsampler (ITU), parameterized by ψ, then maps this prediction to the target-resolution clean latent $\widehat{z}^H_{0|s} = U_\psi(\widetilde{z}^L_{0|s})$.

To continue the original denoising trajectory, InTraScale reconstructs the next target-resolution scheduler state as

$$x^H_{s+1} = (1-\sigma_{s+1})\widehat{z}^H_{0|s} + \sigma_{s+1}\epsilon^H(\xi), \qquad \epsilon^H(\xi) \sim \mathcal{N}(0, I),$$

where $\epsilon^H(\xi)$ is target-grid Gaussian noise deterministically drawn from generation seed ξ. This parameter-free operation constructs a target-resolution state at scheduler level $s+1$, without auxiliary diffusion or a decode–encode round trip. We require $s < T$, so the frozen predictor performs at least one target-resolution step. The remaining target-resolution steps refine structural inconsistencies and synthesize details underdetermined by the low-resolution latent. The complete inference pipeline is

$$x^L_1 \xrightarrow{\text{LR}} x^L_s \xrightarrow{\text{TTD}} \widetilde{z}^L_{0|s} \xrightarrow{\text{ITU}} \widehat{z}^H_{0|s} \xrightarrow{\text{reconstruction}} x^H_{s+1} \xrightarrow{\text{HR}} z^H_{\text{out}}.$$

Here, z^H_{out} denotes the final target-resolution clean latent, which is decoded into the output video by the frozen VAE.

The ITU and TTD parameters are trained independently using supervision generated by the frozen base generator. ITU learns the cross-resolution clean-latent correspondence, whereas the TTD update learns to reduce the unfinished low-resolution trajectory residual at the transition step.

3.3 In-Trajectory Upsampler

The In-Trajectory Upsampler (ITU) addresses the cross-resolution representation mismatch identified in Section 3.1. Rather than treating latent upsampling as spatial interpolation, ITU learns the correspondence between clean low- and target-resolution representations induced by the frozen base generator.

Model-endogenous supervision. We construct the training pairs using videos generated by the frozen base generator. Given a generated target-resolution video v^H, we first obtain its corresponding low-resolution counterpart through RGB-space downsampling:

$$v^L = \mathcal{D}_{\text{rgb}}(v^H).$$

The two videos are then encoded independently by the same frozen VAE $\mathcal{E}$:

$$z^L_{\text{pair}} = \mathcal{E}(v^L), \qquad z^H_{\text{pair}} = \mathcal{E}(v^H).$$

The resulting pair $(z^L_{\text{pair}}, z^H_{\text{pair}})$ shares the same semantics, frame count, and motion trajectory. Performing the resolution change before VAE encoding allows ITU to learn the base generator's actual cross-resolution latent correspondence instead of imitating a predefined interpolation operator in latent space.

Spatiotemporal latent upsampling. ITU maps a clean low-resolution latent to the target-resolution latent grid while preserving the temporal dimension:

$$U_\psi : \mathbb{R}^{B\times 16\times F\times h_L\times w_L} \to \mathbb{R}^{B\times 16\times F\times h_H\times w_H},$$

where B and F denote the batch size and number of latent frames, respectively. The network follows an encoder–resampler–decoder design. A 3D convolution first projects the input into a 256-channel hidden representation, followed by four spatiotemporal residual blocks. A learned spatial lifting operator then changes only the height and width. Four additional residual blocks and a final 3D convolution reconstruct the 16-channel target latent. The 3D blocks allow adjacent latent frames to contribute to the reconstruction without changing the temporal resolution.

We instantiate the spatial lifting operator according to the required grid ratio. For $480 \times 832 \to 720 \times 1248$, the latent grid changes from 60×104 to 90×156. We realize this $3/2$ ratio using a $3\times$ spatial PixelShuffle followed by anti-aliased stride-2 downsampling. For $368 \times 640 \to 720 \times 1248$, a $2\times$ PixelShuffle produces a 92×160 grid, which is center-cropped to 90×156. The resolution-specific lifting operators determine the target geometry, while the surrounding spatiotemporal network learns the latent correspondence.

Cross-resolution objective. For $\widehat{z}^{H}_{\text{pair}} = U_\psi(z^L_{\text{pair}})$, the ITU objective is

$$\mathcal{L}_{\text{ITU}} = \mathcal{L}_{\text{rec}} + 0.2\mathcal{L}_{\text{lf}} + 0.1\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_{\text{rec}}$ is the Charbonnier reconstruction loss, $\mathcal{L}_{\text{lf}}$ enforces consistency between the spatially downsampled prediction and the low-resolution input, and $\mathcal{L}_{\text{temp}}$ matches first-order temporal differences between adjacent latent frames. We use a Charbonnier constant of $\epsilon = 10^{-3}$. Here, $\mathcal{D}_{\text{lat}}$ maps the prediction to the low-resolution grid, and Δ_t denotes first-order temporal differences. The three terms constrain target-latent reconstruction, low-frequency content preservation, and temporal evolution, respectively.

A low-resolution latent does not uniquely determine every high-frequency detail in its target-resolution counterpart. ITU therefore focuses on transferring structure, motion, and latent statistics to the target grid. The remaining frozen target-resolution trajectory retains the generative responsibility for details that are underdetermined by the low-resolution input.

3.4 Trajectory-Tail Distillation

ITU is trained on completed clean-latent pairs, whereas its inference input is predicted from an unfinished low-resolution trajectory. TTD reduces this training–inference mismatch by distilling the omitted low-resolution trajectory tail into a step-specific update. The resulting endpoint-calibrated prediction is subsequently lifted by ITU and returned to the target-resolution continuation.

For endpoint supervision, we run the frozen base generator for each training sample using a fixed prompt, generation seed, initial noise, scheduler, and guidance configuration. We cache the state x^L_s immediately before the transition step and continue the same low-resolution trajectory to completion. We denote the resulting clean endpoint by z^L_{end}. The pair

$$\left(x^L_s, z^L_{\text{end}}\right)$$

therefore preserves the semantic and stochastic conditions at the transition step while providing direct supervision for the trajectory residual remaining at step s.

We implement TTD through a transition-specific low-rank update $\Delta\theta_s$ to the frozen flow predictor. For a linear layer W, the LoRA parameterization is

$$W_\lambda = W + \lambda\frac{\alpha}{r}BA,$$

where r is the LoRA rank, α is its training scale, and λ controls the TTD update strength during inference. We use rank $r = 32$ and inject the update into the q, k, v, and o attention projections and the two feed-forward projections. The base parameters ϕ remain frozen.

At step s, the TTD-augmented predictor produces

$$\widetilde{z}^L_{0|s} = x^L_s - \sigma_s v_{\phi+\lambda\Delta\theta_s}(x^L_s, s, c).$$

We train the TTD parameters to regress the corresponding completed low-resolution endpoint using

$$\mathcal{L}_{\text{TTD}} = \mathcal{L}_1 + 0.1\mathcal{L}_{\text{MSE}} + \lambda_{\text{tt}}\mathcal{L}_{\text{temp}}.$$

Here, $\mathcal{L}_1$ and $\mathcal{L}_{\text{MSE}}$ compare $\widetilde{z}^L_{0|s}$ with z^L_{end}, while $\mathcal{L}_{\text{temp}}$ matches their first-order temporal differences. We set $\lambda_{\text{tt}} = 0.05$ for the earlier transition at step $s = 40$ in the 50-step sampler and zero for the later and distilled settings. We select the inference strength $\lambda = 0.75$ on held-out development prompts and keep it fixed for all reported samples.

The TTD update is applied only at step s. Its strength is zero for the preceding steps, so the frozen low-resolution prefix and the cached pre-transition state x^L_s remain unchanged under matched deterministic conditions. TTD is therefore a step-specificized distillation update rather than a trajectory-wide step-count reduction model.

4 Experiments

We evaluate whether InTraScale improves the quality–efficiency trade-off under conventional 50-step sampling, remains effective with a 4-step timestep-distilled sampler, and addresses the transition failures identified in Section 3.1. We report system-level comparisons followed by component and design analyses.

4.1 Experimental Setup

We evaluate InTraScale on both the standard 50-step Wan2.1 T2V-1.3B sampler and a 4-step Wan2.1 T2V-14B StepDistill-CfgDistill sampler. This section first examines the 50-step setting, where steps 40 and 45 serve as transition steps. We additionally evaluate trilinear interpolation with transition steps 40, 45, and 48 to characterize the effect of increasingly delayed transitions. All configurations generate 81-frame videos at 720×1248 resolution from matched text prompts, random seeds, initial noise, schedulers, and guidance settings.

We evaluate ten matched prompt–seed pairs using VBench. We report subject consistency, background consistency, motion smoothness, aesthetic quality, and imaging quality; Overall (VBench-5) is their unweighted mean, not an official VBench aggregate. Paired analyses use prompt–seed pairs as units.

We measure single-video end-to-end latency after one untimed warm-up, using five matched prompt–seed generations per resident process. CUDA-synchronized timing covers denoising, TTD, ITU, state reconstruction, VAE decoding, and output writing, but excludes runner construction and checkpoint loading.

4.2 Overall Quality–Efficiency Trade-off

We compare InTraScale against Native-HR and three accelerated alternatives. Native-HR executes all 50 steps at the target resolution. Trilinear directly resamples the latent grid at different transition steps. RALU-style combines trilinear transition with noise–timestep rescheduling at step 45. Finally, ITU-5HR (LUVE-style) completes all 50 steps at low resolution, applies ITU to the resulting endpoint, and performs five additional target-resolution refinement steps. Figure 5 visualizes the resulting operating points, while Table 1 reports their exact values.

InTraScale@40 reduces latency from 187.54 to 58.49 seconds, corresponding to a $3.21\times$ speedup, while retaining 97.76% of the Native-HR VBench-5 score. Delaying the

Method	LR/HR	Subj. Cons. ↑	Back. Cons. ↑	Motion ↑	Aesthetic ↑	Imaging ↑	Overall ↑	Latency (s) ↓	Speedup ↑
Native-HR	0/50	0.96140	0.96961	0.98477	0.59493	0.63112	0.82836	187.54	1.00×
Trilinear@40	40/10	0.92179	0.95916	0.97542	0.55312	0.48110	0.77812	58.12	3.23×
InTraScale@40	**40/10**	**0.92713**	**0.95947**	**0.97975**	**0.59451**	**0.58829**	**0.80983**	**58.49**	**3.21×**
Trilinear@45	45/5	0.92263	0.95986	0.97532	0.52675	0.45425	0.76776	41.98	4.47×
RALU-style@45	45/5	0.91940	0.95873	0.97222	0.56949	0.60216	0.80440	42.01	4.46×
InTraScale@45	**45/5**	**0.92359**	**0.95844**	**0.97957**	**0.58032**	**0.59768**	**0.80792**	**42.26**	**4.44×**
Trilinear@48	48/2	0.91995	0.96193	0.97552	0.49471	0.46833	0.76409	32.27	5.81×
ITU-5HR (LUVE-style)	50/5	0.92543	0.95046	0.97542	0.57839	0.57395	0.80073	44.00	4.26×

Table 1: Quality–efficiency comparison for the 50-step sampler. Overall denotes VBench-5.

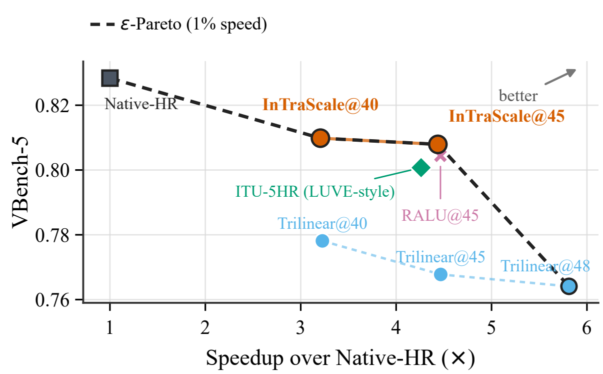

Figure 5: **50-step quality–efficiency Pareto frontier.**

transition to step 45 further reduces latency to 42.26 seconds, achieving a 4.44× speedup and retaining 97.53% of the Native-HR score. Moving the transition from step 40 to 45 reduces latency by an additional 27.74%, with a VBench-5 decrease of only 0.00191. The two operating points therefore provide a controllable quality–efficiency trade-off.

At matched transition positions, the learned transition introduces little additional latency over trilinear interpolation. InTraScale@40 and InTraScale@45 add only 0.63% and 0.67% latency, respectively, while improving VBench-5 by 0.03171 and 0.04016. The widening quality gap at the later transition supports the challenge identified in Section 3.1: as the target-resolution continuation becomes shorter, trilinear interpolation leaves insufficient steps to repair the induced latent distortion.

At step 45, InTraScale also improves VBench-5 over RALU-style rescheduling by 0.00352 with a 0.61% latency increase. This comparison suggests that RALU-style rescheduling combined with trilinear transition does not fully address the VAE-induced cross-resolution correspondence modeled by ITU.

The ITU-5HR (LUVE-style) baseline completes the entire low-resolution trajectory before resolution conversion and adds five target-resolution refinements. Despite executing these additional steps, it reaches only 0.80073 VBench-5 at 44.00 seconds. InTraScale@45 is 3.96% faster and improves VBench-5 by 0.00719. Transitioning before the low-resolution endpoint therefore provides a more favorable operating point than completing low-resolution generation and subsequently launching a separate target-resolution refinement suffix.

Overall, both InTraScale settings are favorable non-dominated operating points on the evaluated tolerance-aware frontier. Their paired confidence intervals against Native-HR overlap zero, but this is not an equivalence test; we therefore do not claim lossless acceleration.

4.3 Further Accelerating Distilled Video Generation

Although Section 4.2 demonstrates substantial acceleration for conventional 50-step sampling, deployment-oriented video generators increasingly rely on timestep-distilled models to reduce the trajectory length. We therefore ask whether in-trajectory resolution scaling can provide additional gains after the step count has already been compressed. To this end, we apply InTraScale to a 4-step Wan2.1 T2V-14B StepDistill-CfgDistill model. Step 3 is the transition step and the last step executed at low resolution, leaving one target-resolution step for generative refinement.

We compare against Native-HR4, which executes all 4 steps at the target resolution, and Trilinear@3, which uses the same 3-LR/1-HR schedule as InTraScale but replaces ITU with fixed trilinear interpolation and disables TTD. We further construct two controlled post-generation baselines on the same distilled backbone. Both complete all four low-resolution steps before increasing the spatial resolution. LUVE-style then applies learned latent upsampling followed by two target-resolution refinements, whereas MrFlow-style uses a decode–RGB-upsample–re-encode path followed by one target-resolution correction. These configurations reproduce the corresponding computational patterns rather than the complete original systems. Figure 6 visualizes these operating points, while Table 2 reports their exact values.

InTraScale-D4@3 reduces latency from 42.49 to 19.92 seconds, achieving a 2.13× speedup while retaining 99.58% of Native-HR4 VBench-5. Its paired confidence interval overlaps zero, we therefore describe this difference as marginal rather than claiming quality equivalence.

Trilinear@3 reaches a slightly higher 2.32× speedup, but its VBench-5 drops by 0.04244 relative to Native-HR4. At the same 3-LR/1-HR schedule, InTraScale adds only 1.59 seconds over Trilinear@3 while improving every component and raising Overall by 0.03884, driven by imaging quality (+0.12635) and aesthetic quality (+0.05960). This result strengthens the observation from the 50-step experiments:

Method	LR/HR	Subj. Cons. ↑	Back. Cons. ↑	Motion ↑	Aesthetic ↑	Imaging ↑	Overall ↑	Latency (s) ↓	Speedup ↑
Native-HR4	0/4	0.97186	0.97557	0.99123	0.65203	0.71135	0.86041	42.49	1.00×
Trilinear@3	3/1	0.95932	0.96937	0.98376	0.58954	0.58783	0.81796	18.33	2.32×
InTraScale-D4@3	**3/1**	**0.96226**	**0.97026**	**0.98815**	**0.64914**	**0.71418**	**0.85680**	**19.92**	**2.13×**
LUVE-style	4/2	0.96238	0.97341	0.98683	0.65191	0.72450	0.85981	28.76	1.48×
MrFlow-style	4/1	0.96379	0.97259	0.98796	0.65491	0.71591	0.85903	27.12	1.57×

Table 2: Quality and efficiency under aggressive 4-step sampling. Overall denotes VBench-5.

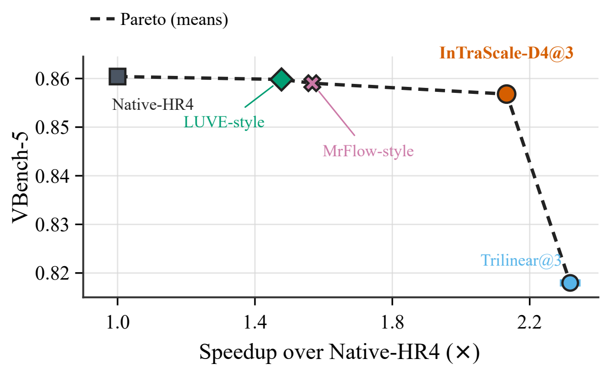

Figure 6: **4-step quality–efficiency Pareto frontier.**

when the target-resolution continuation is reduced to a single step, trilinear interpolation leaves insufficient correction capacity, whereas the learned transition preserves most of the native quality.

InTraScale also provides a more favorable operating point than the post-generation alternatives. It reduces latency by 30.73% relative to LUVE-style and by 26.53% relative to MrFlow-style, while its VBench-5 is lower by only 0.00301 and 0.00223, respectively. The quality difference from MrFlow-style has a prompt-paired confidence interval that overlaps zero; the LUVE-style advantage is small but detectable. InTraScale additionally obtains higher motion smoothness than both post-generation baselines shown in Table 2.

These results demonstrate that in-trajectory resolution scaling is complementary to timestep distillation. Even after the denoising trajectory has been compressed to four steps, InTraScale reduces end-to-end latency by 53.1% while retaining 99.58% of the Native-HR4 VBench-5 score.

4.4 Component and Design Analysis

We next examine whether ITU and TTD address the two transition-specific challenges identified in Section 3.1. On 50 completed clean-latent pairs per scale, ITU consistently improves cross-resolution reconstruction over trilinear interpolation. We compute latent L_1 in latent space and LPIPS/temporal L_1 on decoded RGB frames, with the latter comparing first-order frame differences. For $480\text{p} \rightarrow 720\text{p}$, ITU reduces these three metrics by 48.6%, 73.3%, and 30.8%, respectively. Under the more aggressive $368\text{p} \rightarrow 720\text{p}$ conversion, the corresponding reductions are 33.4%, 51.1%, and 15.5%. These results confirm that ITU learns the VAE-specific cross-resolution correspondence rather than merely approximating geometric interpolation.

TTD consistently brings the transition clean prediction closer to the endpoint of the complete low-resolution trajectory. It reduces endpoint latent L_1 by 25.82%, 21.03%, and 5.03% at the 50-step transitions $s = 40$ and $s = 45$, and the D4 transition $s = 3$, respectively, with improvements on all ten samples in each setting. The factorial evaluation further separates the roles of the two modules: replacing trilinear interpolation with ITU improves VBench-5 by 0.03085–0.04066, whereas enabling TTD changes the aggregate score by only -0.00050 to $+0.00087$. ITU therefore provides the dominant perceptual gain, while TTD primarily reduces the transition-specific endpoint mismatch.

Finally, we validate the target-resolution state reconstruction on an independent eight-prompt set. Using seed-derived target-resolution noise instead of resized low-resolution flow improves VBench-5 by 0.02483 and temporal consistency by 0.01605, showing the importance of reconstructing a scheduler-consistent high-resolution state. Performance remains stable when the TTD update strength varies from 0.25 to 1.0; we select $\lambda = 0.75$ on the validation set and keep it fixed for all test evaluations. Full metric breakdowns and strength sweeps are provided in the supplementary material.

5 Conclusion

We presented InTraScale, which accelerates high-resolution video generation by scaling resolution within a fixed denoising budget, without a separate super-resolution diffusion process. TTD adapts the transition state, while ITU performs VAE-specific cross-resolution latent lifting, allowing the frozen generator to continue at the target resolution. On 50-step Wan2.1 sampling, InTraScale achieves $3.21\times$ and $4.44\times$ speedups while retaining 97.76% and 97.53% of the Native-HR VBench-5 score. On a 4-step distilled model, it achieves $2.13\times$ acceleration with 99.58% score retention. Ablations confirm the complementary roles of TTD and ITU, establishing in-trajectory resolution scaling as a spatial acceleration strategy complementary to timestep distillation.

Limitations. Model-endogenous self-distillation specializes ITU to the VAE and resolution pair and TTD to the transition step. This enables close adaptation without external paired data, but new configurations may require renewed supervision and module adaptation. Future work will explore cross-configuration transfer and content-aware transition policies.

References

Anagnostidis, S.; Bachmann, G.; Kim, Y.; Kohler, J.; Georgopoulos, M.; Sanakoyeu, A.; Du, Y.; Pumarola, A.; Thabet, A.; and Schönfeld, E. 2025. FlexiDiT: Your Diffusion Transformer Can Easily Generate High-Quality Samples with Less Compute. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 28316–28326.

Blattmann, A.; Rombach, R.; Ling, H.; Dockhorn, T.; Kim, S. W.; Fidler, S.; and Kreis, K. 2023. Align Your Latents: High-Resolution Video Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Ding, Z.; Jin, C.; Liu, D.; Zheng, H.; Singh, K. K.; Zhang, Q.; Kang, Y.; Lin, Z.; and Liu, Y. 2025. DOLLAR: Few-Step Video Generation via Distillation and Latent Reward Optimization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 17961–17971.

Gu, J.; Zhai, S.; Zhang, Y.; Susskind, J.; and Jaitly, N. 2024. Matryoshka Diffusion Models. In *International Conference on Learning Representations*, 30234–30262.

He, J.; Xue, T.; Liu, D.; Lin, X.; Gao, P.; Lin, D.; Qiao, Y.; Ouyang, W.; and Liu, Z. 2024. VEnhancer: Generative Space-Time Enhancement for Video Generation. arXiv:2407.07667.

Ho, J.; Chan, W.; Saharia, C.; Whang, J.; Gao, R.; Gritsenko, A.; Kingma, D. P.; Poole, B.; Norouzi, M.; Fleet, D. J.; and Salimans, T. 2022. Imagen Video: High Definition Video Generation with Diffusion Models. arXiv:2210.02303.

Jeong, W.; Lee, K.; Seo, H.; and Chun, S. Y. 2025. Training-Free Mixed-Resolution Latent Upsampling for Spatially Accelerated Diffusion Transformers. arXiv:2507.08422.

Jin, Y.; Sun, Z.; Li, N.; Xu, K.; Xu, K.; Jiang, H.; Zhuang, N.; Huang, Q.; Song, Y.; Mu, Y.; and Lin, Z. 2025. Pyramidal Flow Matching for Efficient Video Generative Modeling. arXiv:2410.05954.

Kim, D.; Ghadiyaram, D.; and Gadde, R. 2026. DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 11459–11471.

Li, J.; Feng, W.; Fu, T.-J.; Wang, X.; Basu, S.; Chen, W.; and Wang, W. Y. 2024. T2V-Turbo: Breaking the Quality Bottleneck of Video Consistency Model with Mixed Reward Feedback. arXiv:2405.18750.

Li, S.; Barnes, C.; Rizve, M. N.; Peng, H.; Li, Z.; Dibua, O.; Ganjdanesh, A.; Tang, X.; Kang, Y.; and Gong, Y. 2026. Content-Aware Dynamic Patchification for Efficient Video Diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 35936–35945.

Liu, F.; Zhang, S.; Wang, X.; Wei, Y.; Qiu, H.; Zhao, Y.; Zhang, Y.; Ye, Q.; and Wan, F. 2025. Timestep Embedding Tells: It's Time to Cache for Video Diffusion Model. arXiv:2411.19108.

ModelTC. 2026. LightX2V: Variable Resolution Inference and Step Distillation Documentation. https://github.com/ModelTC/LightX2V. Accessed: 2026-07-15.

Nie, W.; Berner, J.; Ma, N.; Liu, C.; Xie, S.; and Vahdat, A. 2026. Transition Matching Distillation for Fast Video Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4645–4655.

Wan, T.; Wang, A.; Ai, B.; Wen, B.; Mao, C.; Xie, C.-W.; Chen, D.; Yu, F.; Zhao, H.; Yang, J.; Zeng, J.; Wang, J.; Zhang, J.; Zhou, J.; Wang, J.; Chen, J.; Zhu, K.; Zhao, K.; Yan, K.; Huang, L.; Feng, M.; Zhang, N.; Li, P.; Wu, P.; Chu, R.; Feng, R.; Zhang, S.; Sun, S.; Fang, T.; Wang, T.; Gui, T.; Weng, T.; Shen, T.; Lin, W.; Wang, W.; Wang, W.; Zhou, W.; Wang, W.; Shen, W.; Yu, W.; Shi, X.; Huang, X.; Xu, X.; Kou, Y.; Lv, Y.; Li, Y.; Liu, Y.; Wang, Y.; Zhang, Y.; Huang, Y.; Li, Y.; Wu, Y.; Liu, Y.; Pan, Y.; Zheng, Y.; Hong, Y.; Shi, Y.; Feng, Y.; Jiang, Z.; Han, Z.; Wu, Z.-F.; and Liu, Z. 2025. Wan: Open and Advanced Large-Scale Video Generative Models. arXiv:2503.20314.

Wang, X.; Zhang, S.; Zhang, H.; Liu, Y.; Zhang, Y.; Gao, C.; and Sang, N. 2023a. VideoLCM: Video Latent Consistency Model. arXiv:2312.09109.

Wang, Y.; Chen, X.; Ma, X.; Zhou, S.; Huang, Z.; Wang, Y.; Yang, C.; He, Y.; Yu, J.; Yang, P.; Guo, Y.; Wu, T.; Si, C.; Jiang, Y.; Chen, C.; Loy, C. C.; Dai, B.; Lin, D.; Qiao, Y.; and Liu, Z. 2023b. LaVie: High-Quality Video Generation with Cascaded Latent Diffusion Models. arXiv:2309.15103.

Xi, H.; Yang, S.; Zhao, Y.; Xu, C.; Li, M.; Li, X.; Lin, Y.; Cai, H.; Zhang, J.; Li, D.; Chen, J.; Stoica, I.; Keutzer, K.; and Han, S. 2025. Sparse Video-Gen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 68208–68224.

Xiao, H.; Chao, B.; Yariv, L.; and Wetzstein, G. 2026. Spectral Progressive Diffusion for Efficient Image and Video Generation. arXiv:2605.18736.

Xie, L.; Li, Y.; Du, S.; Xia, M.; Wang, X.; Yu, F.; Chen, Z.; Wan, P.; Zhou, J.; and Dong, C. 2025. SimpleGVR: A Simple Baseline for Latent-Cascaded Video Super-Resolution. arXiv:2506.19838.

Xu, Y.; Park, T.; Zhang, R.; Zhou, Y.; Shechtman, E.; Liu, F.; Huang, J.-B.; and Liu, D. 2024. VideoGigaGAN: Towards Detail-Rich Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhao, C.; Chen, J.; Li, H.; Kang, Z.; Lu, S.; Wei, X.; Zhang, K.; Yang, J.; and Tai, Y. 2026. LUVE: Latent-Cascaded Ultra-High-Resolution Video Generation with Dual Frequency Experts. arXiv:2602.11564.

Zhao, T.; Fang, T.; Huang, H.; Wan, R.; Soedarmadji, W.; Liu, E.; Li, S.; Lin, Z.; Dai, G.; Yan, S.; Yang, H.; Ning, X.; and Wang, Y. 2025a. ViDiT-Q: Efficient and Accurate Quantization of Diffusion Transformers for Image and Video Generation. In *International Conference on Learning Representations*.

Zhao, X.; Jin, X.; Wang, K.; and You, Y. 2025b. Real-Time Video Generation with Pyramid Attention Broadcast. arXiv:2408.12588.

Zheng, S.; Chen, G.; He, L.; Liu, J.; Lin, Y.; Zou, C.; and Zhang, L. 2026a. From Sketch to Fresco: Efficient Diffusion Transformer with Progressive Resolution. arXiv:2601.07462.

Zheng, X.; Liu, X.; Ding, Y.; Feng, W.; Lin, J.; Guo, J.; and Qin, H. 2026b. Multi-Resolution Flow Matching: Training-Free Diffusion Acceleration via Staged Sampling. arXiv:2607.01642.

Zhou, S.; Yang, P.; Wang, J.; Luo, Y.; and Loy, C. C. 2024. Upscale-A-Video: Temporal-Consistent Diffusion Model for Real-World Video Super-Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhuang, J.; Guo, S.; Cai, X.; Li, X.; Liu, Y.; Yuan, C.; and Xue, T. 2026. FlashVSR: Towards Real-time Diffusion-Based Streaming Video Super Resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 43482–43493.

Zou, C.; Li, C.; Liu, S.; Zhong, Z.; Huang, K.; and Zhang, L. 2026. DisCa: Accelerating Video Diffusion Transformers with Distillation-Compatible Learnable Feature Caching. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 4590–4601.